

BeyondGrip: Physics Informed Control Recovery for Race Cars

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Motivation



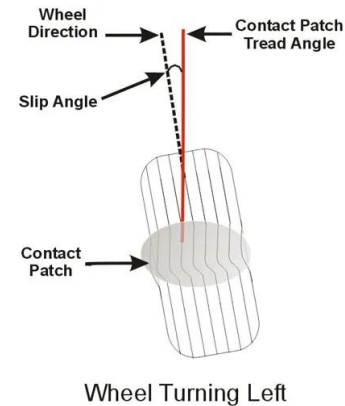
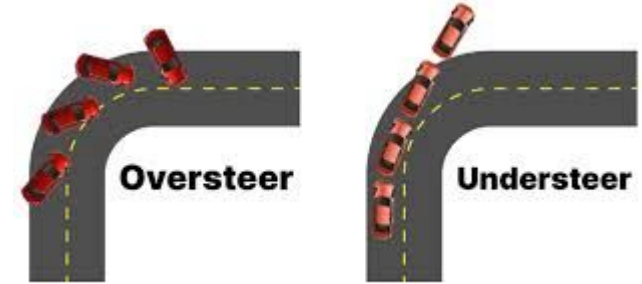
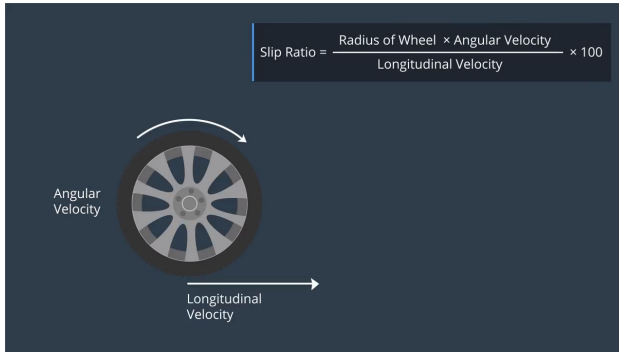
- Loss of vehicle control can happen anytime in daily driving due to rain, snow, or black ice, often with no warning and very little time to react.
- In both public roads and racing at the limit, fast and safe recovery is critical to prevent accidents and reduce loss of performance.
- As opposed to standard Recovery RL, we want to learn to return not only to any safe state, rather we want to return to sufficiently good safe states.

Objective

- Develop an RL-based controller to regain vehicle stability during sudden loss of control, focusing on oversteer scenarios.
- Enable fast and safe recovery by learning adaptive corrective actions that handle nonlinear tire dynamics and varying grip conditions.
- Embed physics knowledge through physics-informed rewards to improve robustness and generalization to real-world driving and racing conditions.
- Evaluate and compare RL algorithms (TD3, SAC, physics-informed TD3) in Assetto Corsa Gym using custom testing scenarios, assessing recovery speed, stability, and reliability.
- Study hierarchical policy design with relative and absolute action-policies.

Background (Terminology)

- **Oversteer:** Rear of the car slides out, vehicle tends to spin.
- **Understeer:** Front of the car slides, vehicle resists turning.
- **Slip Angle:** Angle between the wheel's actual path and the direction it's pointing.
- **Slip Ratio:** Relative difference between wheel speed and vehicle speed.



Simulation Setup - Assetto Corsa

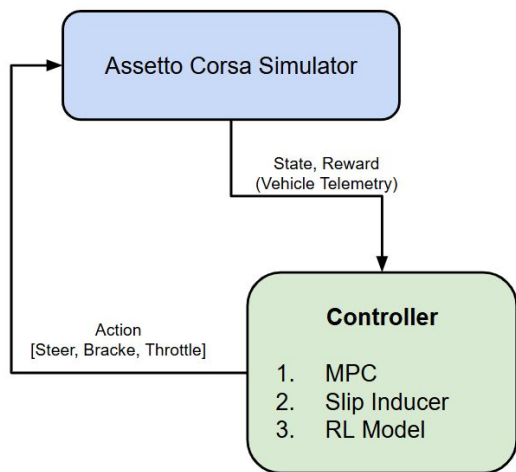
Why Assetto Corsa for Slip Recovery?

- High-fidelity physics-based racing simulator
- Realistic tire dynamics & slip behavior
- Rich vehicle telemetry:
 - Velocity, yaw, pitch, roll
 - Slip angle, slip ratio, per-wheel forces & load
- Supports extreme driving regimes
 - Oversteer, understeer, loss of traction

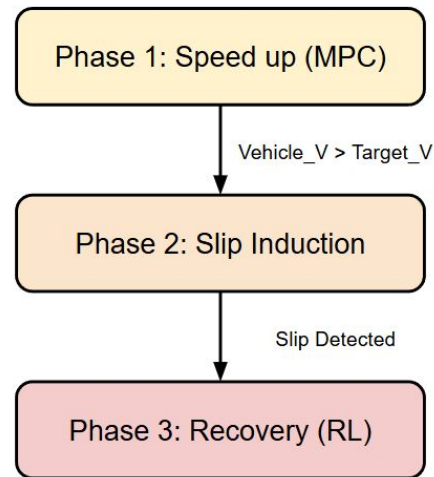


Simulation Setup - Assetto Corsa

RL Control & Slip Scenario



- Standard RL interaction loop with continuous vehicle control.
- Controller outputs steering, brake, and throttle actions.



- Episode are divided into speed-up, slip, and recovery phases.
- Phase transition are condition-based.

Methodology

Slip Induction & Episode Initialization

- MPC-based initialization
 - Vehicle driven along reference trajectory to a target speed.
- Slip induction via steering flick
 - Short counter-directional steering impulse used to induce loss of rear traction.
- Randomized disturbance parameters
 - Flick direction, magnitude, and duration are sampled within bounded ranges.
- Control handoff to RL agent
 - Control transferred when slip angle reaches 5–20°.
- Focused training regime
 - Agent trained only on recoverable but unstable vehicle states.

Reward Design (Dense Rewards)

Following Reference Line:

$reward += 6.0 * clip(gap_to_line / 0.5)$

Alive Bonus:

$reward += 1.9 + 3.0 * stability$

Slip Penalty:

$reward -= (0.8 * (slip_angle / 5.0)^2) * time_decay$

Gap Penalty:

$reward -= (0.1 * (gap_to_line / 2.0)^2) * time_decay$

Final Rewards:

$R_total = 0.2 * clip(R, -8.0, 2.0)$

Reward Design (Terminal Rewards)

Reference line: Pre-computed racing line

If *Recovered*:

- Speed: Faster end speed -> Higher rewards
- Gap to reference line: Smaller gap to reference line -> Higher rewards
- Time: Shorter recovery time -> Higher rewards

If *Failed (Car leave track or crash)*:

- Speed: Faster end speed -> Higher penalty
- Gap to reference line: Bigger gap to reference line -> Higher penalty
- Time: Shorter time to failure -> Higher penalty

Reward Design (Physics Rewards)

Enable Condition: Rear Slip Angle $> 3^\circ$ (Oversteer is happening)

Load Transfer:

Oversteer $\rightarrow$ less rear grip.

We want more rear load $\rightarrow$ more rear grip $\rightarrow$ recover from oversteer.

Slip Ratio Control:

More rear load $\rightarrow$ strong acceleration $\rightarrow$ high slip ratio $\rightarrow$ can reduce rear grip.

We want lower slip ratio $\rightarrow$ rewards for keeping lower slip ratio.

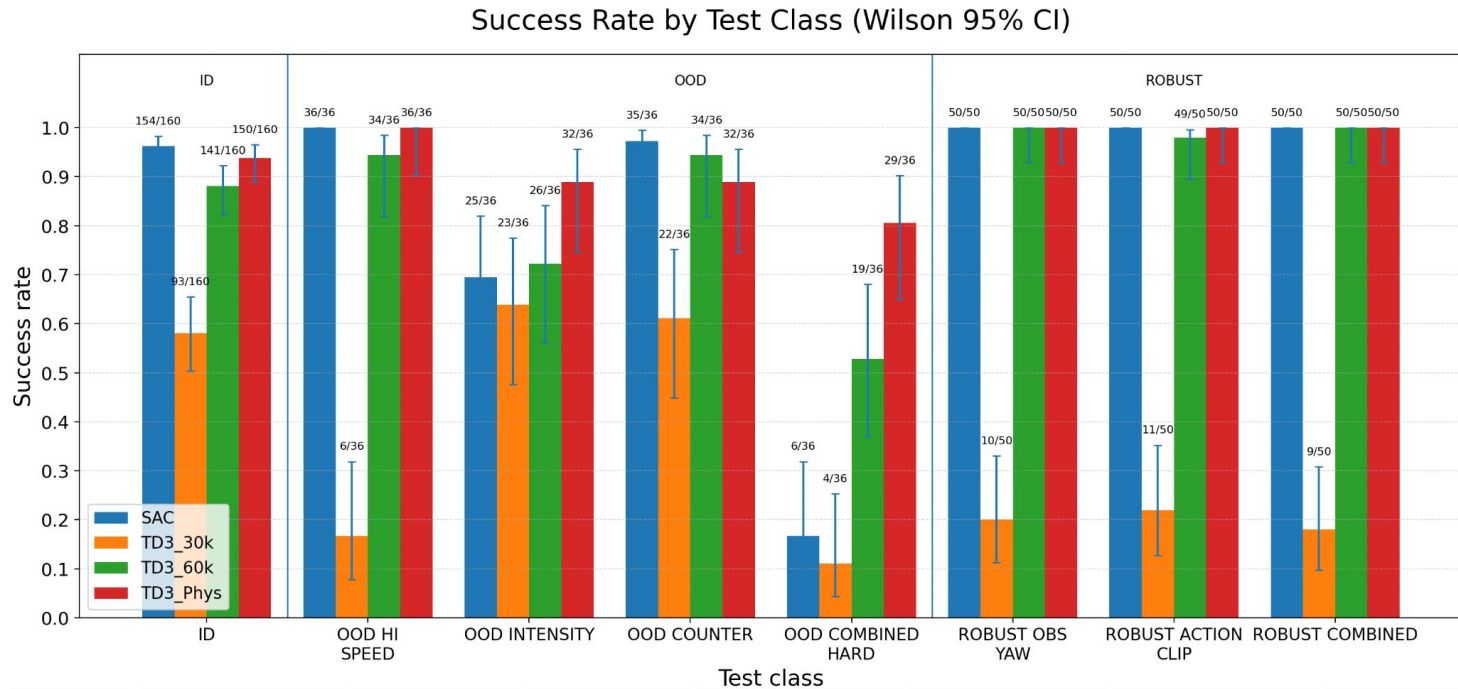
Counter Steering:

Target steering angle is calculated based on rear slip angle. More points for correct counter steering.

Steering Reaction:

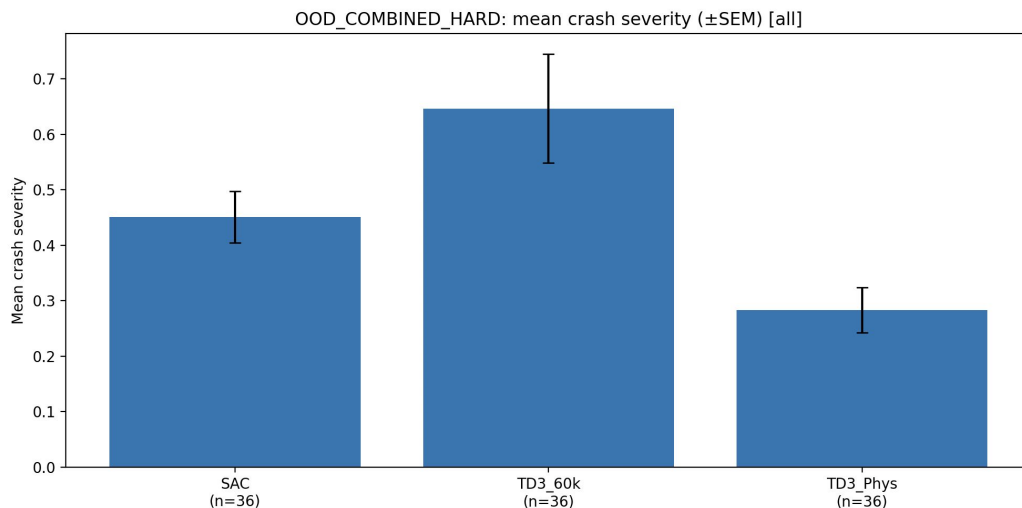
Steering velocity is calculated by self alignment torque to prevent causing another oversteer by holding counter steering too long.

Experiment



Experiment

- OOD Test
 - +10% higher max speed
 - 2x more intense flick
 - 50% higher counter-steer magnitude
- Severity of crashes for different algorithms (lower is better)
- We focus on the hard case where baseline TD3 fails in the OOD test
- TD3 Physics algorithm recovers is ~60% of cases



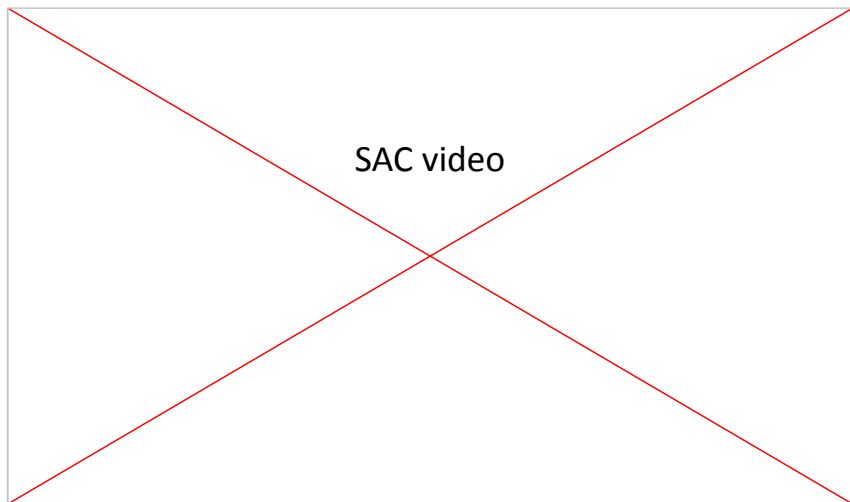
Demo

Slip Event Case



- Slip-induced scenario (no recovery)

Slip Recovery Result



- SAC using non-physics reward



- TD3 with physics-informed reward

Next Step

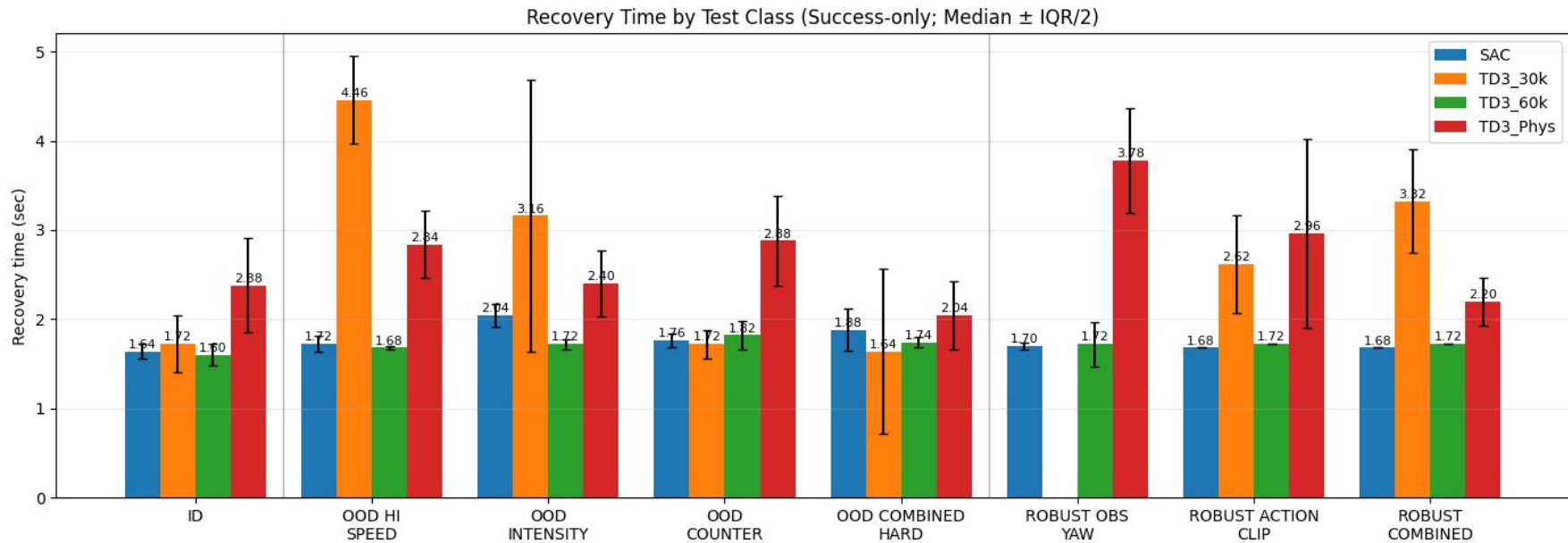
- Expand training beyond straight-line oversteer.
- Include more varied scenarios
- Formalize transitions into and out of the recovery policy

Bonus

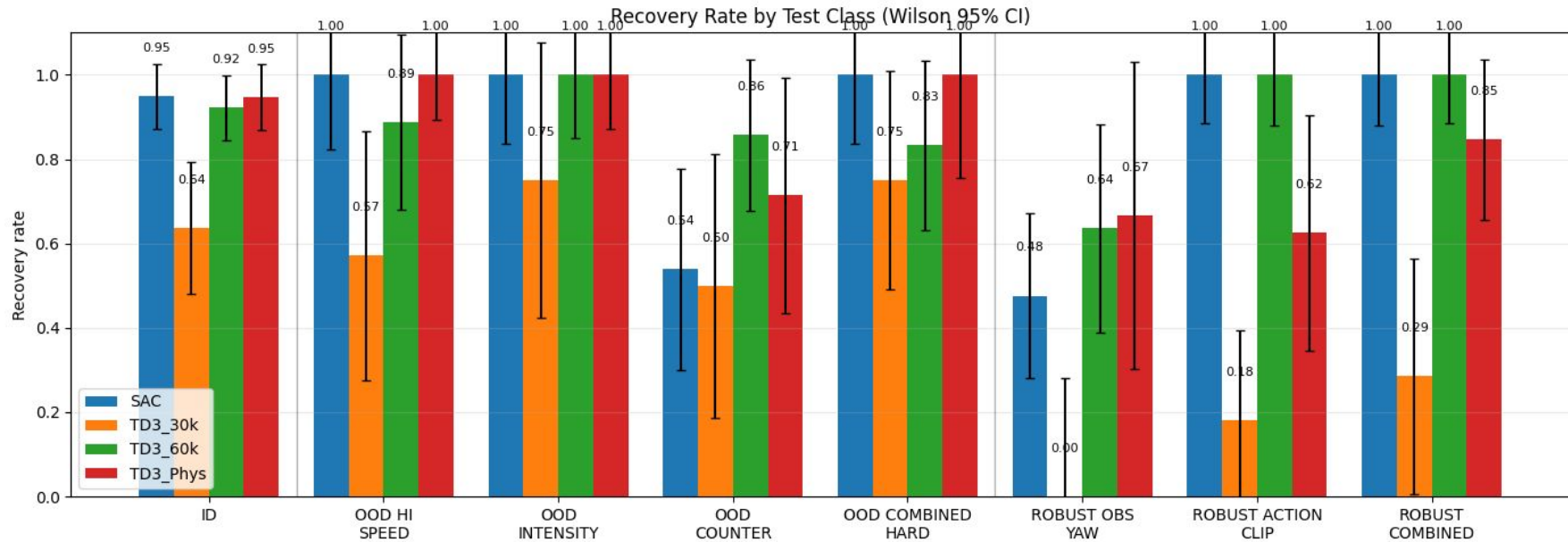


Thank you!

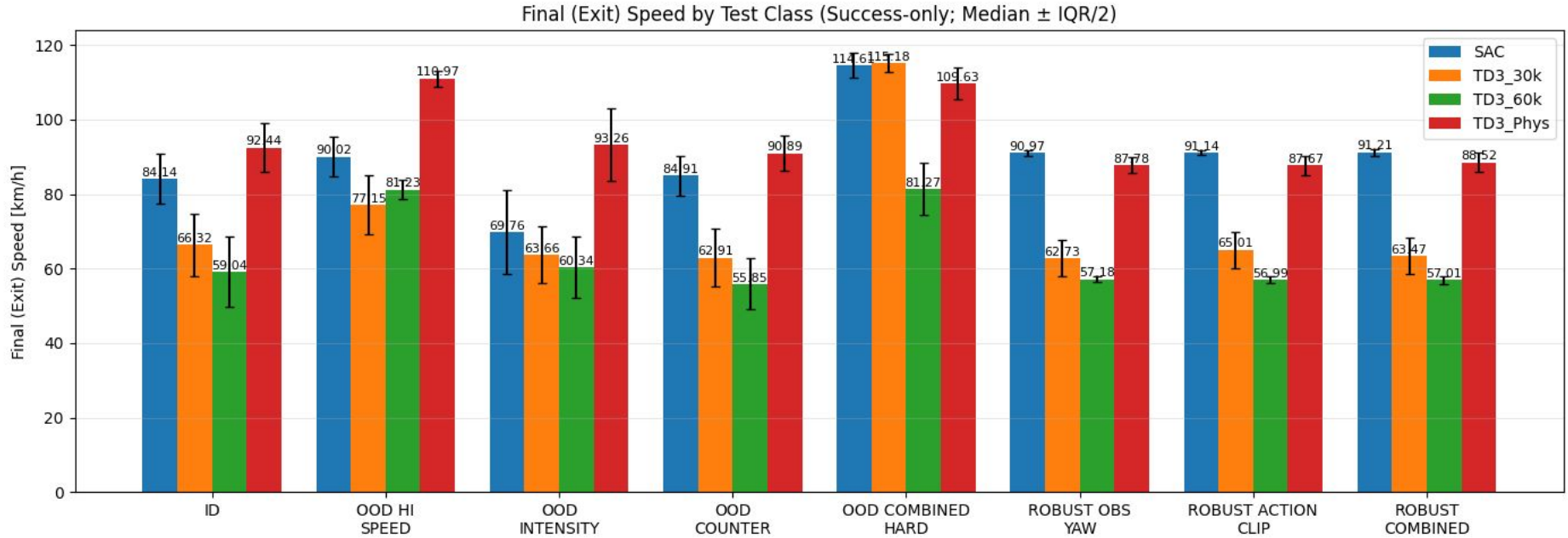
Experiment - Recovery Time



Experiment - Recovery Rate



Experiment - Exit (final) Speed



Background (Previous Research)

- Assetto Corsa Gym was developed as a high-fidelity simulation benchmark for autonomous racing, aimed at enabling safe and scalable training of autonomous agents.
- Researchers used large-scale human driving data to model and evaluate autonomous driving policies.
- SAC (Soft Actor-Critic) was employed to compare learned policies against human driving performance.
- Demonstrated the platform's capability for benchmarking RL algorithms in complex racing scenarios and evaluating policy generalization.

Remonda, A., Hansen, N., Raji, A., Musiu, N., Bertogna, M., Veas, E., & Wang, X. (2024). A simulation benchmark for autonomous racing with large-scale human data. *Advances in Neural Information Processing Systems*, 37, 102078-102100.

State and Action Space Design

State

- Vehicle kinematics: speed, acceleration, velocity
- Slip indicators: per-wheel slip angles
- Orientation: yaw, pitch, roll
- Physics-informed signals: tire grip, wheel load

Action

- Steering: $[-1, +1]$
- Throttle, brake: $[0, 1]$
- Absolute action formulation for rapid slip correction